



SINA OJAGHI

Sabbatical PhD Students

DoB: 19/9/1995
Marital Status: Single
Military Service: Educational Exemption

As a Ph.D. student in Computer Architecture at the University of Tehran, my academic journey has been deeply committed to research and education. Having served as an instructor at both Hamedan University of Technology and the University of Tehran, I've contributed significantly to my students' academic growth while honing my teaching and research skills. Currently, I balance my roles as IT Manager at HamVta Arya Co. and a hardware programmer at Afra Turbine Energy Co., applying my technical expertise to tackle real-world challenges. At my core, a passion for discovery and innovation propels me forward. My extensive involvement in research and development projects underscores my dedication to advancing computer architecture. I'm now seeking a study opportunity to enhance my research capabilities further, aiming to offer groundbreaking insights into hardware security and the applications of spiking neural networks in sustainable technologies

Contact

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Education

PhD in Computer Engineering

Branch: Computer Architecture
Tehran University
Tehran, Iran
GPA: 18.05
2020 - Present

Master of Computer Engineering

Branch: Computer Architecture
Islamic Azad University Science and Research Branch
Tehran, Iran
GPA: 18.45
2017 - 2019

Bachelor of Computer Engineering

Branch: Computer Architecture
Hamadan University of Technology
Hamadan, Iran
GPA: 16.34
2013 - 2017

Work Experience Academic

Research Assistant

Institute for Research in Fundamental Sciences (IPM)
Tehran, Iran
September 2024 - Present

Tasks and Achievements :

- Research Assistant at the Institute for Research in Fundamental Sciences (IPM), conducting research under the supervision of Dr. Ahmad Khonsari and Dr. Kamyar Givaki.

Research Assistant

University of Tehran
Tehran, Iran
April 2024 - Present

Tasks and Achievements :

- Research Assistant at the University of Tehran, focusing on FPGA hardware security, side-channel attacks, and PUFs under Dr. Siamak Mohammadi.

Instructor Network Laboratory

University of Tehran
Tehran, Iran
September 2024 - Present

Tasks and Achievements :

- Computer Networks Laboratory Instructor at the University of Tehran, teaching undergraduate computer networks laboratory courses and supervising hands-on networking experiments.

Instructor Artificial Intelligence (AI)

Hamedan University of Technology
Hamadan, Iran
September 2021 - October 2023

Tasks and Achievements :

- Department of Computer and IT Engineering and Department of Electrical Control and Electronics Engineering

Instructor Research Methods (RM)

Hamedan University of Technology
Hamadan, Iran
September 2021 - October 2023

Tasks and Achievements :

- Department of Electrical Control and Electronics Engineering and Department of Computer and IT Engineering

Secretary and speaker of Cyber-Physical Systems society if iran

CPSSI
Tehran, Iran
March 2022 - Present

Tasks and Achievements :

- link: <https://cpsci.ir/>

Idea Developer and Senior Consultant of Targeted Internship Site

Hamedan University of Technology
Hamadan, Iran
September 2016 - January 2019

Tasks and Achievements :

- link: <http://intern.hut.ac.ir/>

Language

English

Score 513 on the Tolimo language test
(2023)

Reading 

Writing 

Speaking 

Listening 

Persian

Native speaker

Reading 

Writing 

Speaking 

Listening 

Social Network

 [linkedin.com/in/sina-ojaghi](https://www.linkedin.com/in/sina-ojaghi)

 [researchgate.net/profile/Sina-Ojaghi](https://www.researchgate.net/profile/Sina-Ojaghi)

 join.skype.com/invite/HFvbHgQg7dM3

 t.me/sinaoj

 github.com/sinaojaghi

Honors

- 13th place among the candidates in the 2020 PhD entrance exam for Computer Architecture.
- 1st place in the M.S. class of 2017, Computer Architecture, Islamic Azad University, Science and Research Branch.
- 2nd place in the B.S. class of 2013, Computer Architecture, Hamedan University of Technology.
- 1st place in the ACM Competitions, Hamedan province

Certificates

The Top project of the computer department of Hamadan University of Technology to enter the Elcamp exhibition in Hamadan province (2016)

Institute: Hamadan University of Technology

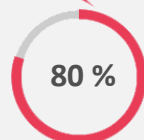
The Top project of the computer department of Hamadan University of Technology to enter the exhibition of research, technology and market technology of Hamadan province (2016)

Institute: Hamadan University of Technology

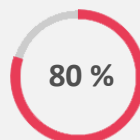
The Top project of the Computer Department of Hamadan University of Technology Smartening of smart hospitals to lead to a conference on the role of economic prosperity in the prevention of social crimes (2017)

Institute: Hamadan University of Technology

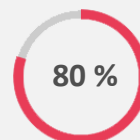
Skills



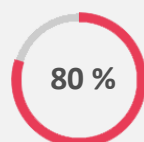
VHDL Programming
Language



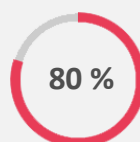
Matlab Programming
Language



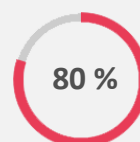
Python Programming



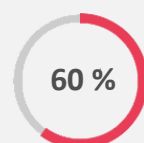
Embedded linux
Raspberry Pi



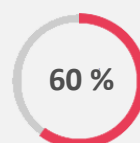
System Verilog



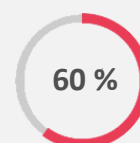
Arm Programming



Altium Designer



Programming Language
C++, C , Java



Cisco CCNA

Design and Implementation of Remote Side-Channel Attacks and Countermeasures on Spiking Neural Networks

Publisher : University of Tehran

Ph.D. Proposal

Design and manufacture of audio processors for smartening domestically produced vehicles

Publisher : Islamic Azad University, Science and Research Branch

Master Thesis

Transfer voice data under secure protocol and use it in smart homes

Publisher : Hamedan university of technology

Bachelor Thesis

Ojaghi, Sina, et al. "Evaluating the Impact of Active Wire Fence Placement on Remote Power Side-Channel Attacks in Multi-Tenant FPGAs." 2026 Fourth International Conference on Distributed Computing and High Performance Computing (DCHPC). IEEE, 2026.

Link : <https://ieeexplore.ieee.org/abstract/document/11517259/>

Maleki, Pouria, Abbas Ramazani, Hassan Khotanlou, Sina Ojaghi, Milad Mousavi, Alexey Kalinin, and Amir Mosavi. "Object Detection for Vehicles with Yolo". 2024 IEEE 22nd World Symposium on Applied Machine Intelligence and Informatics (SAMI): IEEE, 2024.

Link : <https://ieeexplore.ieee.org/document/10432884>

Maleki, Pouria, Abbas Ramazani, Hassan Khotanlou, and Sina Ojaghi. "Iranian Vehicle Images Dataset for Object Detection Algorithm." Journal of AI and Data Mining 12, no. 1 (2024): 127-36.

Link : https://jad.shahroodut.ac.ir/article_3164.html

Ojaghi, Sina, Farzan Barati, and Somaye Jafari jassbi. "Stress-Strain Modeling of an Al-Mg Using an Artificial Neural Network." Paper presented at the 3rd International Conference on Electrical, Computer and Mechanical Engineering papers, 2020.

Link : <https://civilica.com/doc/1128854/>

Ojaghi, Sina, Farzan Barati, and Somaye Jafari jassbi. "Modeling of Az52 Flow Curve at High Temperature by Implicating the Genetic Algorithm Method." Paper presented at the 3rd International Conference on Electrical, Computer and Mechanical Engineering papers, 2020.

Link : <https://civilica.com/doc/1128855/>

Ojaghi, Sina, and Javid Ghafourian. "Offline voice detection in smart homes." The 4th International Conference on Electrical Engineering, Computer, Mechanics and Artificial Intelligence. 2023.

Link : <https://en.civilica.com/doc/1780938/>

Projects

Ideation, Design and Implementation of Targeted Internship Site

For : Hamadan University of Technology

2016

The site we designed worked for about 3 years at the university and then became inaccessible due to the more comprehensive Golestan system.

Link : <http://intern.hut.ac.ir>

Verification Calculator II & III & RISC-V

For : Under Supervision : Dr. Siamak Mohammadi (University of Tehran)

January 2021

This verification includes: circuit performance simulation, design of a Stimuli production platform, Golden Model design, Checker design, Scoreboard design and implementation of Coverage techniques

Training and replacing the heavy part of the circuit with a neural network and accelerating the circuit

For : Under Supervision : Dr. Mehdi Modaresi (University of Tehran)

June 2021

Chip multiprocessor (CMP) Advanced lesson project

Investigation and detection of hardware malware in AES-T300 RS232-T400

For : Under Supervision : Dr. Siamak Mohammadi (University of Tehran)

June 2021

Hardware Security and Trust Advanced lesson project

Automatic annotation of photos with genetic algorithm in MATLAB environment

For : Under Supervision : Dr. Mohsen Fathiyan (Hamadan University of Technology)

2017

computer vision Advanced lesson project and this project was used to identify the status of the signature and the owner of the signature in banking operations.

Data transfer in FPGAs (UART) in VHDL language in ISE environment

For : Under Supervision : Dr. Mohsen Ahmad Vand (Hamadan University of Technology)

2015

Data transfer Advanced lesson project

MIPS processor design with some commands in VHDL language in ISE environment

For : Under Supervision : Dr. Ehsan Aerabi (Hamadan University of Technology)

2009

Computer architecture Advanced lesson project

Work Experience az T.A. and ACS and School

Teaching Assistant for Master's and PhD Programs at University of Tehran

■ Hardware Security and Trust

Fall 2022, Fall 2023

■ Functional Verification of HDL Models

Fall 2021, Fall 2022

■ Performance Evaluation

Spring 2022, Spring 2023, Spring 2024, Spring 2025

■ Computer Networks (B.Sc.)

Fall 2021, Fall 2022, Fall 2023, Fall 2024, Fall 2025

Teaching Assistant for Bachelor Programs at Hamedan University of Technology

■ Microprocessor and Assembly Language

September 2015 - June 2016

■ Computer Architecture

December 2015 - December 2016

■ Logic Circuits

December 2015 - June 2016

■ Discrete Mathematics

December 2015 - June 2016

Teaching for Association of Computer Science at Hamedan University of Technology

■ Data Transfer and Telecommunications

December 2016 - June 2017

■ Microprocessor and Assembly Language

September 2016 - June 2017

■ Computer Architecture

February 2015 - January 2017

■ Digital Electronics

December 2016 - June 2017

■ Logic Circuits

September 2014 - December 2015

■ Algorithm Design

December 2015 - June 2016

■ C Programming Language

September 2014 - December 2015

Teacher

■ IOT

School Khadije Kobra Tehran

Tehran, Iran

September 2017 - May 2019

■ Computer

School Dar ol Fonoon Hamedan

Hamedan, Iran

September 2014 - May 2016

Idea Developer and Senior Consultant of Targeted Internship Site

Hamedan University of Technology
Hamedan, Iran September 2016 - January 2019

Tasks and Achievements :

- link: <http://intern.hut.ac.ir/>

Hardware Programmer

Afra Turbine Energy Co.
Hamedan, Iran September 2022 - Present

Tasks and Achievements:

- Designed and implemented embedded software for microcontroller-based hardware systems
- Developed firmware updates to improve the functionality and performance of hardware devices

IT Manager

HamVeta Co.
Tehran, Iran September 2023 - June 2024

Tasks and Achievements :

- Developing and implementing IT policies and procedures to streamline processes and enhance security

R & D

Nabz
Tehran, Iran November 2019 - April 2020

Tasks and Achievements :

- link: <https://nabzgroup.com/>

Supporter, Group Manager

Qalam Chi Educational Foundation
Hamadan, Iran
September 2015 - May 2017

Laptop and Computer Software and Hardware Repairs

DPC
Hamadan, Iran
May 2013 - September 2014

References

Dr. Siamak Mohammadi

Computer department group manager Tehran University
smohammadi@ut.ac.ir

Dr. Ahmad Khonsari

Member of the Scientific Council of the IPM Basic Knowledge Research Institute
ak@ipm.ir

Dr. Somaye Jafar Ali Jasbi

Computer department group manager Islamic Azad university science and research branch
s.jassbi@srbiau.ac.ir

Dr. Ahmad Khadem Zade

Deputy of Iran Telecommunication Research Center and member of the Scientific Council of Islamic Azad University, Science and Research Branch
a_khademzadeh@azad.ac.ir

Dr. Ehsan Aerabi

PhD student of Grenoble University of Technology and former faculty of Hamadan University of Technology
e_aerabi@comp.iust.ac.ir

Dr. Mir Hossin Dezfoulial

Computer department group manager Bu ali sina university
dezfoulial@basu.ac.ir

